Role of Variable Domain in the binding of Dynamin-related Protein 1 (Drp1) to its adaptor, Mitochondrial Fission Factor (Mff)

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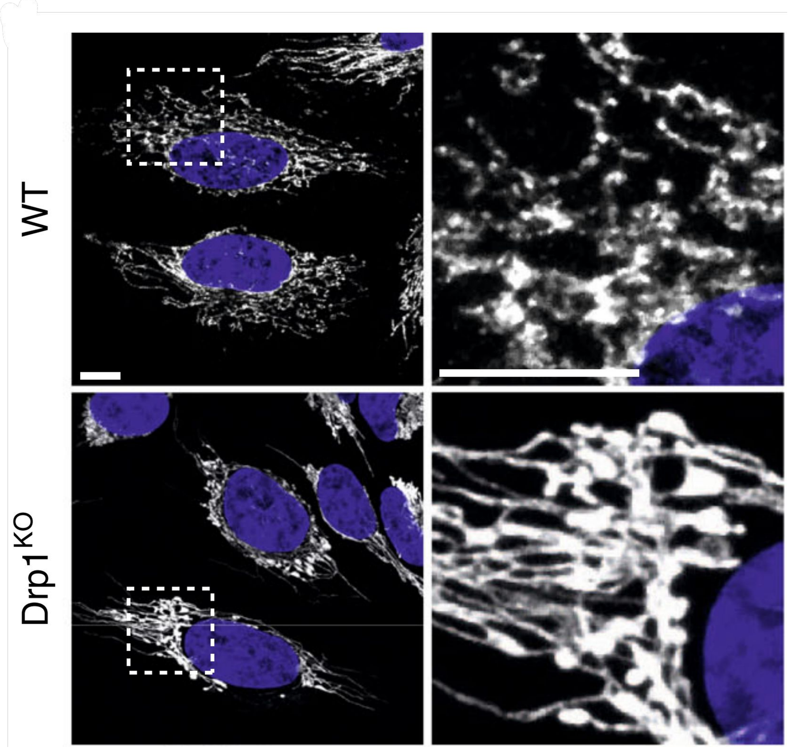
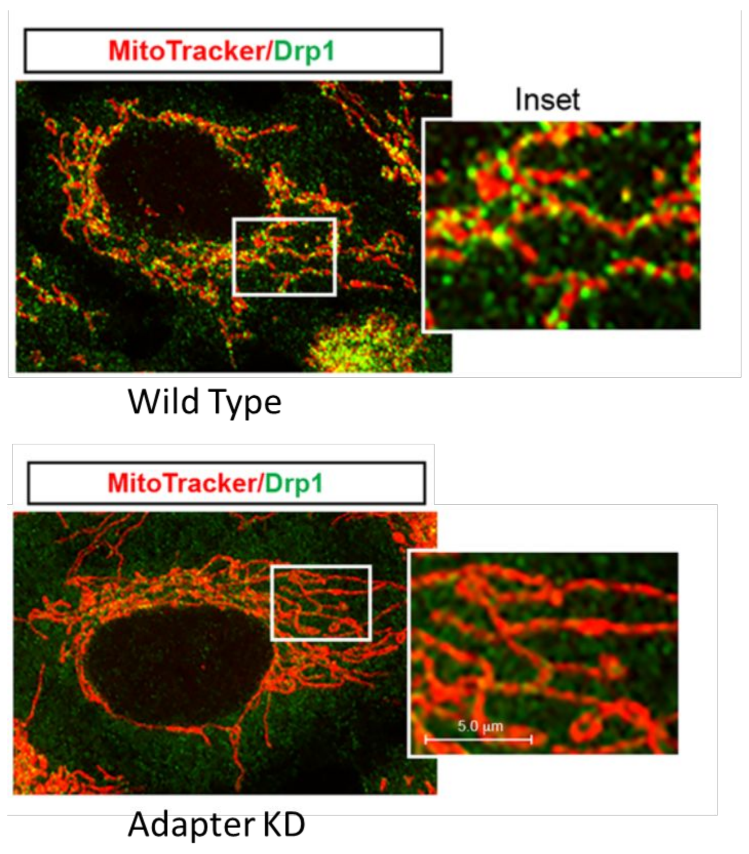
Introduction

- Mitochondria are dynamic organelles that undergo constant fusion and fission
- Fission of mitochondria is regulated by a protein called Drp1, a large GTPase that belongs to the dynamin superfamily
- Drp1's recruitment to mitochondria depends on adaptor proteins – Mitochondrial Fission Factor (Mff), MiD49/51, Fis1
- Knock down of either Drp1 or adaptor proteins makes mitochondria tubular and elongated

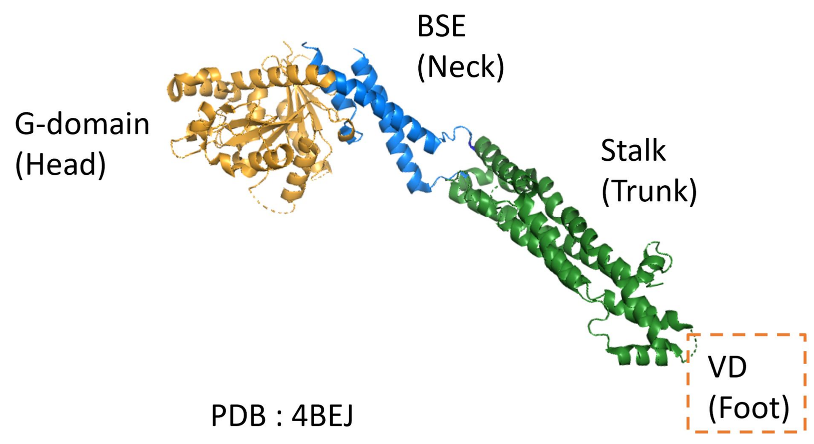


Kamerkar *et al.* (2018)

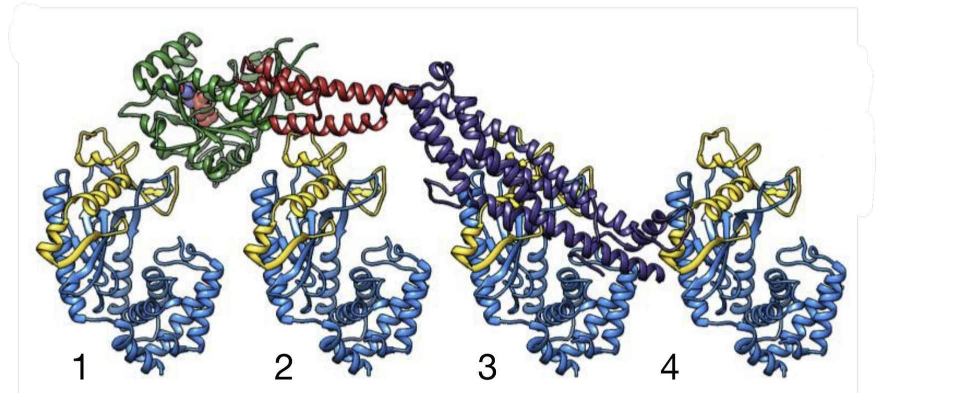


Yu *et al.* (2017)

- Structural basis of Mid49 interaction with Drp1 has been reported before

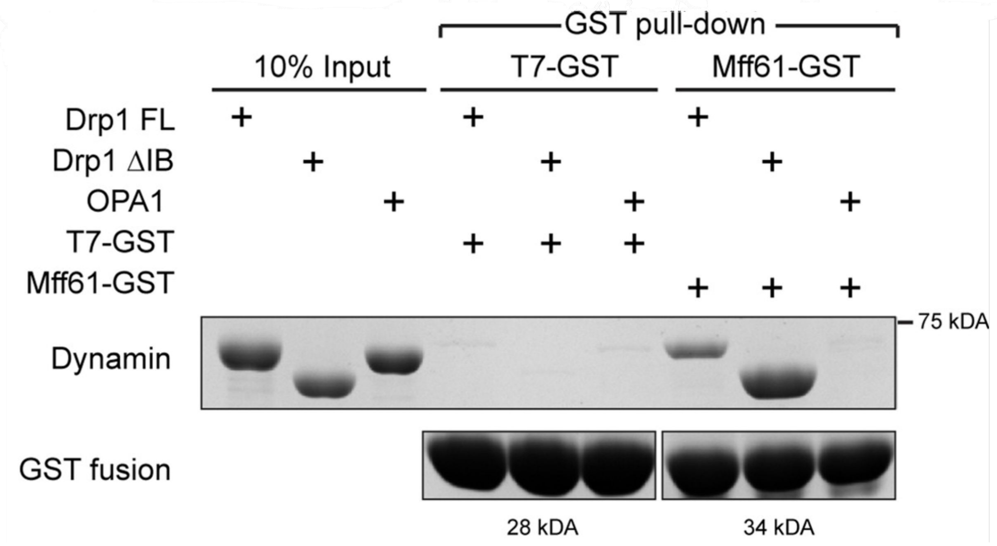


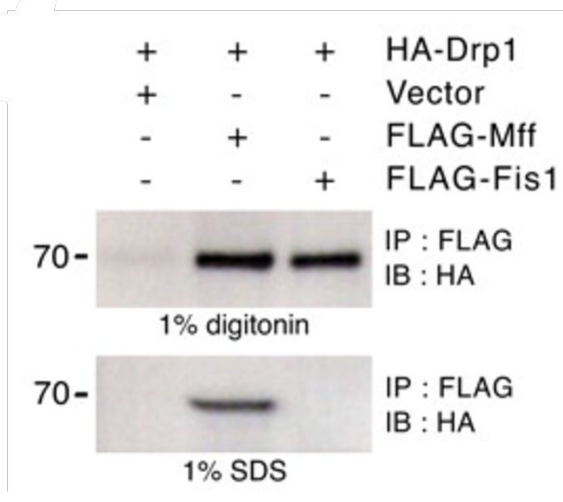
Fröhlich *et al.* (2013)



Kalia *et al.* (2018)

- It has been reported that Mff doesn't interact with Drp1 strongly but immunoprecipitation followed by crosslinking showed that it can Mff can interact with Drp1

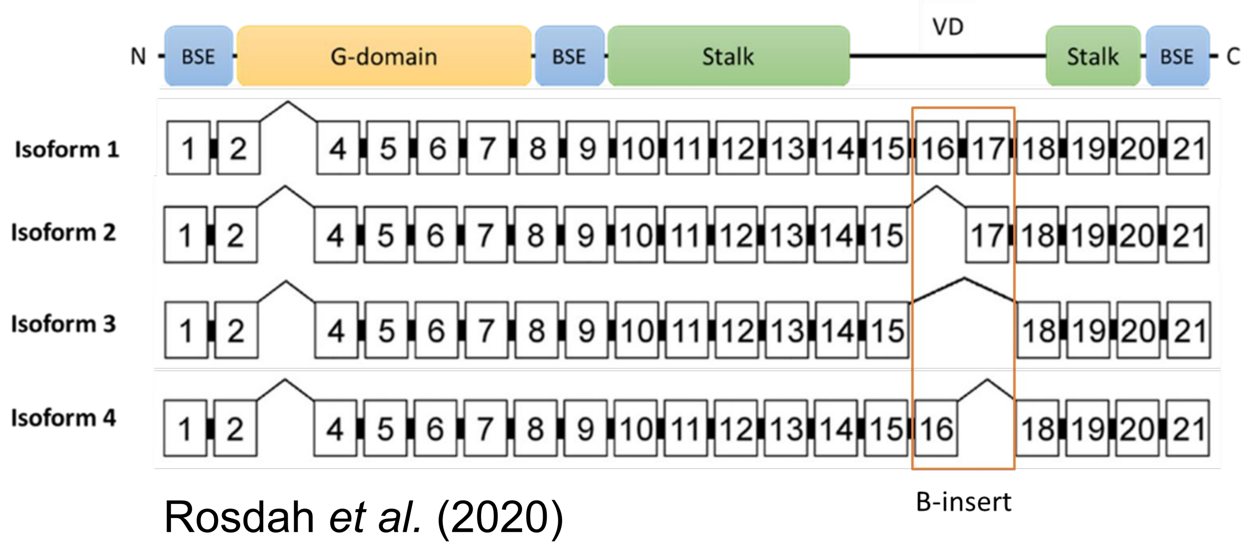




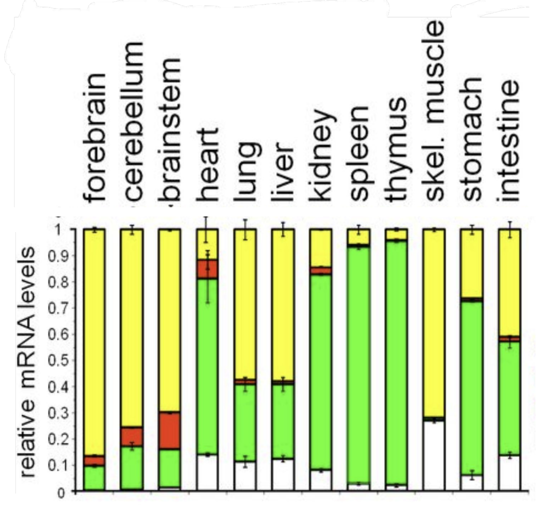
Liu *et al.* (2015)
Otera *et al.* (2010)

Alternative splicing of Drp1 makes isoforms of Drp1 with VD of variable length

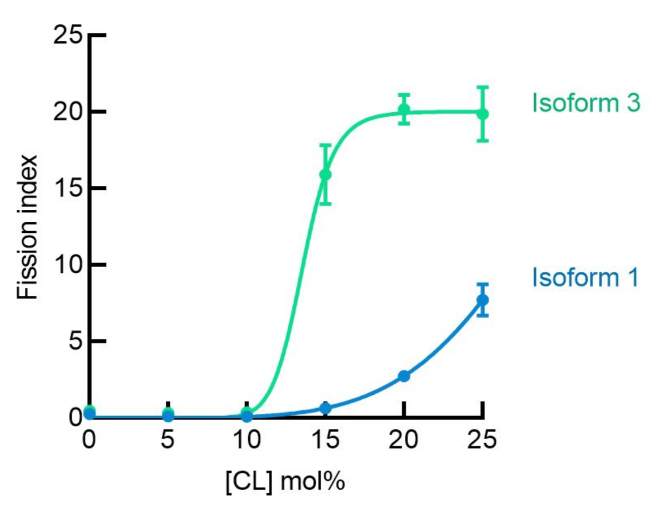
- Drp1 has four naturally occurring isoforms that have variable length of V-domain
- These isoforms show tissue-specific expression
- These isoforms show differential fission activity on membrane nanotubes
- These isoforms also show differential rescue in mitochondrial morphology when expressed in Drp1 KO cell (data not shown)



Rosdah *et al.* (2020)

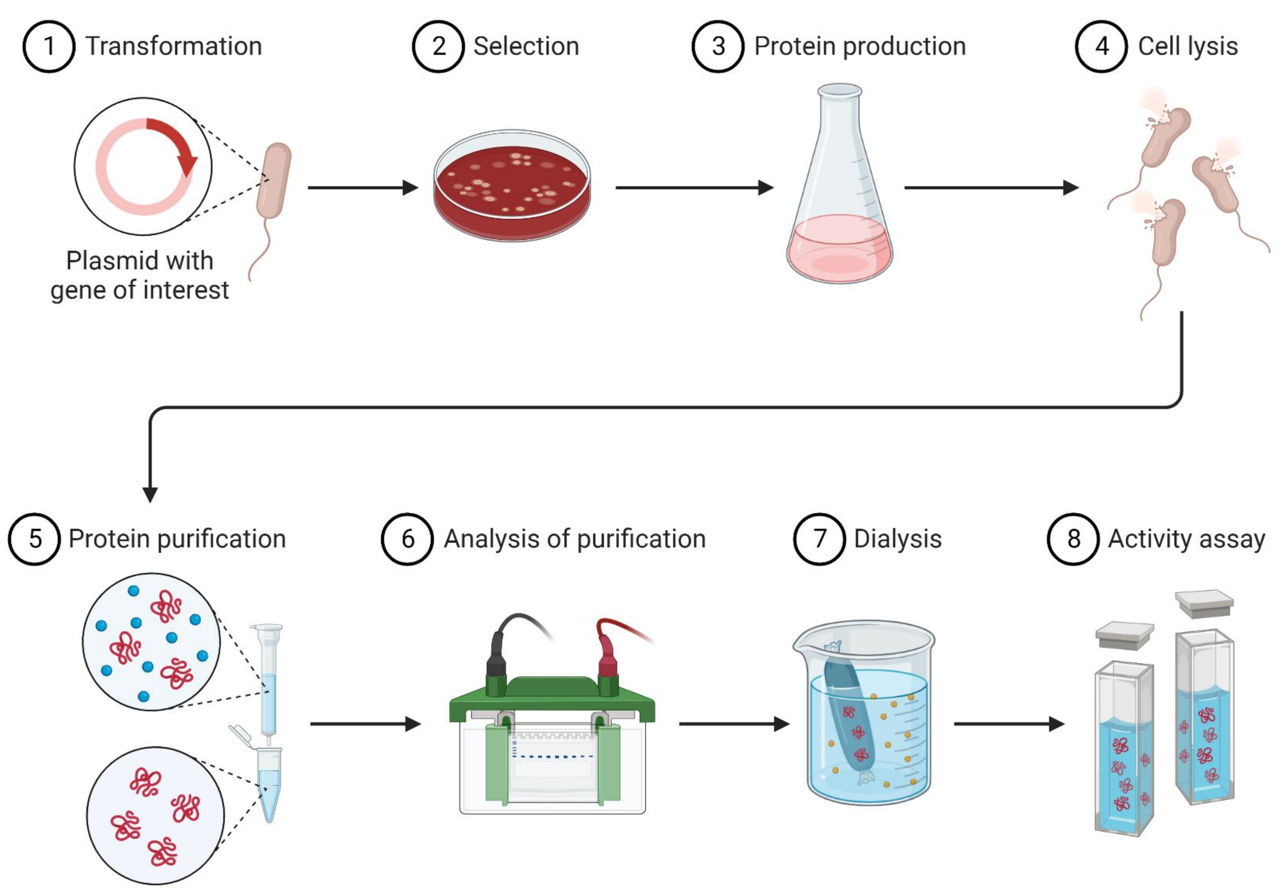


Strack *et al.* (2013)

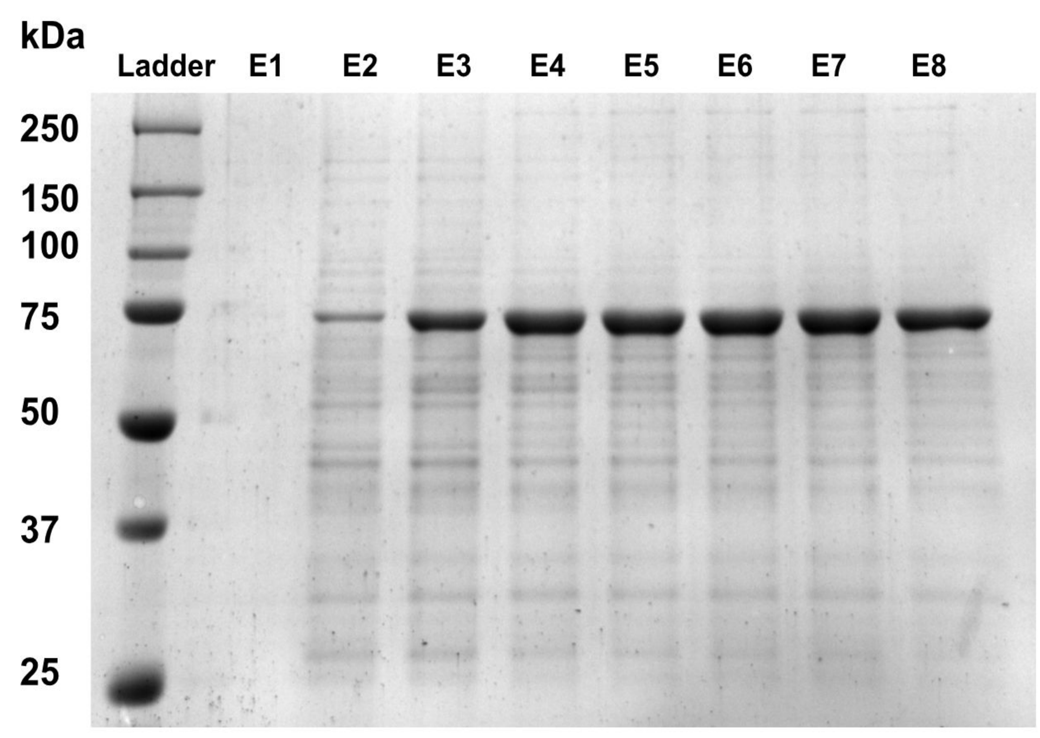


De, Pucadyil unpublished data

Protein expression and purification

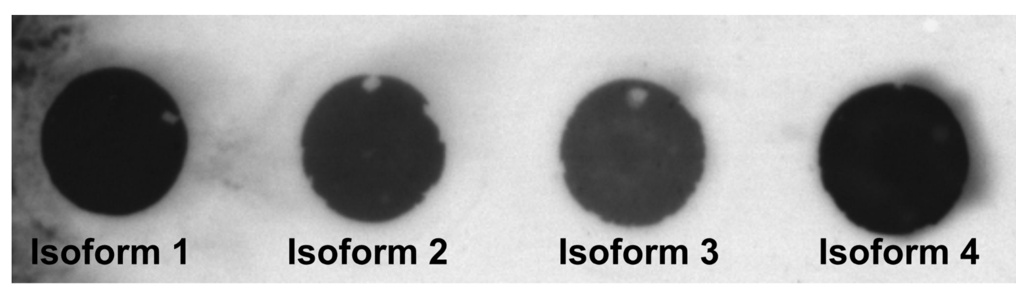


Four isoforms of Drp1 tagged with StrepII affinity tag at the C-terminus and Mff-mCherry tagged with His at the C terminus and StrepII at the N terminus were purified using affinity chromatography, and purity was checked by running SDS-PAGE

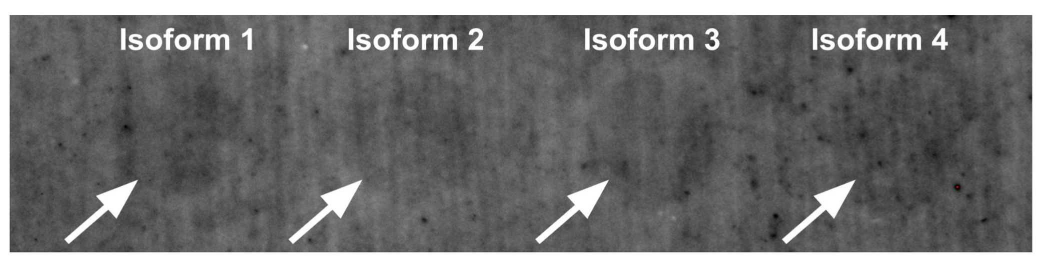


SDS-PAGE gel depicts purified Drp1 Isoform 3

Checking for protein-protein interactions



Imaged PVDF membrane when Ponceau S dye was added to the membrane loaded with the bait proteins (Isoforms 1-4)



Membrane obtained from performing dot-blot displaying very faint signal of binding on the membrane attributable to protein-protein interactions between Drp1 isoforms and Mff-mCherry

Discussion

- We show that Drp1 isoforms are incapable of effectively binding to Mff on its own
- We hypothesize that Drp1's interaction with Mff might be affected by the presence of cardiolipin (CL), an essential phospholipid that makes up to 10% of the outer mitochondrial membrane
- *In vitro* fission experiment on membrane nanotubes has shown that Drp1's fission activity can be governed by coincident detection of both Mff and CL (Kamerkar *et al.* 2018)
- This is in accordance with work from other labs that showed Drp1's enzymatic activity doesn't get affected by only the presence of Mff (Clinton *et al.* 2016)

Acknowledgement

I am grateful to my mentor, Sannidhya De's patient guidance throughout the course of the project and to Prof. Pucadyil for providing me with the opportunity to work in his lab. I would also like to thank other members of the Pucadyil Lab for their timely guidance and support.

References

1.Kamerkar, S.C., Kraus, F., Sharpe, A.J. et al. Dynamin-related protein 1 has membrane constricting and severing abilities sufficient for mitochondrial and peroxisomal fission. Nat Commun 9, 5239 (2018). https://doi.org/10.1038/s41467-018-07543-w
2.Otera H, Wang C, Cleland MM, et al. Mff is an essential factor for mitochondrial recruitment of Drp1 during mitochondrial fission in mammalian cells. J Cell Biol. 2010;191(6):1141-1158. doi:10.1083/jcb.201007152
3.Rosdah AA, Smiles WJ, Oakhill JS, et al. New perspectives on the role of Drp1 isoforms in regulating mitochondrial pathophysiology. Pharmacol Ther. 2020;213:107594. doi:10.1016/j.pharmthera.2020.107594
4.Francy CA, Clinton RW, Fröhlich C, Murphy C, Mears JA. Cryo-EM Studies of Drp1 Reveal Cardiolipin Interactions that Activate the Helical Oligomer. Sci Rep. 2017;7(1):10744. Published 2017 Sep 6. doi:10.1038/s41598-017-11008-3